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SIC-AI Capstone Project

Mid-Term Progress Report

Speaker-Aware Noise Cancellation and Target Speech Enhancement

Submitted By

Austina Aryal (079BEI010)
Prashamsa Aryal (079BEI024)
Sostika Shrestha (079BEI041)

Submitted To

Arun Timilsena and Suresh Pokharel
Supervisors, SIC-AI Capstone Project

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Abstract

This report presents the research, design, implementation, and evaluation of our SIC-AI capstone project on target speaker extraction (TSE). The objective is to extract the voice of a specified speaker from a single-channel speech mixture containing interfering speakers and background noise, using a short enrollment recording of the target speaker as a reference.

We implemented and trained a speaker-conditioned Band-Split Recurrent Neural Network (BSRNN) integrated with a frozen ECAPA-TDNN speaker encoder. The ECAPA-TDNN extracts a 192-dimensional speaker embedding from the enrollment audio, which is incorporated into the separator through speaker-feature addition and FiLM conditioning across eight dual-path layers. The resulting model contains approximately 16.83 million parameters. It was trained on dynamically generated mixtures of LibriSpeech speech and DEMAND environmental noise using a combined scale-invariant signal-to-distortion ratio (SI-SDR) and multi-resolution STFT loss.

The model completed 100 training epochs, achieving its best validation SI-SDR improvement (SI-SDRi) of **11.84 dB at epoch 97**, evaluated on unseen speakers. On the 1,200-mixture test set, the epoch-97 checkpoint achieved an SI-SDR improvement of **11.6 dB** on non-clean mixtures and increased STOI from **0.77 to 0.88**, indicating improved speech intelligibility. A Whisper-based evaluation also showed a reduction in word error rate from **38.3% to 18.1%**. The model selected the incorrect speaker in **7.7% of interferer-containing mixtures**, with most errors occurring when enrollment recordings were short.

These results demonstrate that the implemented speaker-conditioned BSRNN can effectively extract a target speaker from mixtures containing competing speech and environmental noise. The report also discusses the system's limitations, including its non-causal architecture, and outlines remaining work involving conditioning ablation, evaluation on unseen noise conditions, and CPU inference measurements.

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1 Introduction

1.1 Problem and motivation

Conventional noise suppression removes environmental noise but treats every human voice as signal. In a call or a meeting, a second person talking nearby is not noise in that sense, yet is still unwanted. Target speaker extraction (TSE) handles this by conditioning the enhancement network on an enrollment recording of the desired speaker, so that only that voice is kept [1, 2]. Because the output is tied to one enrolled speaker, TSE avoids the permutation problem of blind speech separation and does not need to know how many speakers are present.

1.2 Objectives

The main objective of this project is to design, implement, and evaluate a deep-learning-based, speaker-conditioned target-speaker extraction system that extracts a target speaker's speech from single-channel mixtures containing interfering speakers and background noise, using a short enrollment recording as a speaker reference.

The specific objectives are as follows:

1. To design and implement a target-speaker extraction system that integrates a pretrained ECAPA-TDNN speaker encoder with a speaker-conditioned Band-Split Recurrent Neural Network (BSRNN) separator;
2. To train and evaluate the proposed system on speech mixtures containing competing speakers and environmental noise, assessing extraction performance using objective metrics such as SI-SDR improvement (SI-SDRi), STOI, and word error rate (WER);
3. To investigate the system's generalization and robustness by evaluating its performance on unseen speakers and noise conditions, and examining target-speaker selection errors under different enrollment conditions;
4. To assess the model's computational requirements and practical limitations, including parameter count, model size, inference efficiency, and potential deployment on resource-constrained systems;
5. To develop a reproducible training and evaluation framework that supports dataset preparation, mixture generation, model training, checkpointing, and systematic performance evaluation.

1.3 Scope of this Report

This report documents the work completed against the proposed project timeline, the design changes made during implementation and the rationale behind them, the current system architecture and dataset, the results obtained to date, and the remaining work and associated risks.

The model is non-causal and processes the input using both past and future temporal context. Therefore, it is not designed for real-time, streaming speech extraction and may introduce processing latency. In addition, when the target speaker's voice is substantially quieter than an interfering speaker, the extracted output may contain residual interference or speech leakage from the interfering speaker. Performance may also be affected by the quality and duration of the enrollment recording.

2 Progress and Changes from the Proposal

Table 1 compares the work packages in the proposal timeline with their current status.

Table 1: Status of the proposal work packages at mid-term.

Work package	Status	Notes
Literature review and dataset survey	**Done**	Reviewed in the proposal and extended with BSRNN and dual-path work (Section 3).
Audio pipeline and classical baseline	**Partial**	STFT/iSTFT pipeline done. The classical baseline was not evaluated; no classical baseline has been implemented.
Dataset preparation and mixture generation	**Done**	LibriSpeech and DEMAND manifests, fixed validation and test sets, on-the-fly scene sampler (Section 5).
ECAPA-TDNN encoder and conditioning ablation	**Partial**	Frozen ECAPA-TDNN with cached embeddings done; additive and FiLM conditioning implemented. The concat/FiLM/gating ablation is not done.
Separator backbones	**Partial**	Conv-TasNet reached 5.90 dB validation SI-SDRi at epoch 21 before being dropped; speaker-conditioned BSRNN trained for 100 epochs (Section 2). TF-GridNet not attempted.
Curriculum training and augmentation	**Partial**	Randomized scenarios, SIR/SNR and levels. No curriculum schedule and no reverberation yet.
Evaluation and generalization tests	**In progress**	Test metrics, scenario/SNR/SIR breakdowns, speaker confusion and WER are documented (Section 7); ablations and unseen-noise tests are pending.
Latency, CPU and memory optimization	**Partial**	GPU cost measured offline (RTF 0.028, 16.83 M parameters). CPU speed and causal latency are not measured; the model is non-causal.
Final validation and documentation	**Pending**	This report is the first documentation milestone.

The main achievements so far are: a speaker-conditioned BSRNN with 16.83 M parameters, trained for the full 100 epochs, with a best validation SI-SDRi of 11.84 dB at epoch 97 on unseen speakers; a test evaluation of the epoch-97 checkpoint with an 11.6 dB SI-SDR gain, STOI rising from 0.77 to 0.88 and PESQ rising from 1.33 to 2.38 on non-clean mixtures; a Whisper word error rate falling from 38.3% to 18.1%; a measured speaker-confusion rate of 7.7%; and a resumable training pipeline with cached embeddings, fixed manifests and an inference path that accepts a user's own mixture and enrollment recording.

2.1 Design changes

The largest change was the separator backbone. The proposal put Conv-TasNet first and BSRNN second. We trained a causal Conv-TasNet with $N = 512$, $H = 512$, film2 conditioning, 3 s segments, batch 3 with gradient accumulation 2, and learning rate 10^{-3} for a planned 80 epochs. The saved best checkpoint is epoch 21 with validation SI-SDRi 5.90 dB; a 20-clip check gave 3.88 dB on training voices and 3.60 dB on held-out voices and noise. We dropped it because BSRNN reached 11.84 dB validation SI-SDRi; Table 16 lists the baseline configuration. Table 2 lists the other deviations.

3 Background and Related Work

Target speaker extraction. VoiceFilter [1] conditions a spectrogram mask estimator on a d-vector of the target speaker. SpeakerBeam-SS [2] uses lightweight Conv-TasNet variants with state-space modeling for low-latency operation. We use the same enrollment-conditioned formulation but a band-split time–frequency backbone.

Table 2: Differences between the proposal and the current implementation.

Aspect	Proposal	Current	Reason
Separator	Causal Conv-TasNet; BSRNN as a stretch goal	Speaker-conditioned BSRNN, 8 dual-path layers	Conv-TasNet reached 5.90 dB validation SI-SDRi, below BSRNN's 11.84 dB
Causality	Strictly causal (past and current frames)	Non-causal (bidirectional LSTMs)	Causal operation is deferred
STFT	512-point FFT, 160-sample hop (10 ms)	512-point FFT, 128-sample hop (8 ms)	Finer time resolution for BSRNN
Speaker encoder	ECAPA-TDNN, frozen versus fine-tuned	Frozen ECAPA-TDNN, cached embeddings	Saves GPU time and memory
Data	VoiceBank, DEMAND, MUSAN, DNS	LibriSpeech (1172 training speakers) and 16 DEMAND environments	Many clean speakers; one noise source
Reverberation	Limited RIR simulation	Not included yet	Listed as future work
Loss	SI-SDR + spectral + mask terms	SI-SDR (clamped at 30 dB) + 5× multi-resolution STFT	Waveform and spectral objective; mask term dropped
Test protocol	Speaker- and noise-disjoint sets	Speaker-disjoint; noise audio disjoint, environments shared	See Section 8

Band-split RNN. BSRNN [3] splits the complex spectrogram into sub-bands, projects each band to a common feature size and alternates RNNs along time and along bands, in the spirit of dual-path models [4]. It was first used for music separation and later for efficient speech enhancement [5]. TF-GridNet [6] is a heavier alternative that we keep as a possible upper-bound reference.

Speaker embeddings and conditioning. ECAPA-TDNN [7] maps an utterance to a fixed-length speaker embedding; we use the SpeechBrain implementation [8]. Feature-wise linear modulation (FiLM) [9] scales and shifts feature channels from a conditioning vector and is our main speaker-injection mechanism.

Losses and metrics. We optimize scale-invariant SDR [10] with a multi-resolution STFT loss [11], and evaluate with SI-SDRi, STOI [12] and wideband PESQ [13]. The DNS Challenge [14] and DeepFilterNet [15] motivated the original efficiency goals.

4 Current System Design

4.1 Overview

Given a mixture $x = s_t + s_i + n$ (target, interferer, noise) and an enrollment embedding $\mathbf{e} \in \mathbb{R}^{192}$ of the target, the network predicts a complex mask M on the mixture spectrogram X and returns $\hat{s}_t = \mathrm{iSTFT}(M \odot X)$. Figure 1 shows the pipeline.

5 Data and Mixture Simulation

LibriSpeech [16] supplies clean read English at 16 kHz. We train on train-clean-100 and train-clean-360 (251 + 921 = 1172 speakers), validate on dev-clean (40 speakers) and test on test-clean (40 speakers). Validation and test speakers never appear in training, so these sets measure generalization to unseen voices. The validation set is 300 fixed mixtures (seed 1234) and the test set is 1200 fixed mixtures (seed 4321).

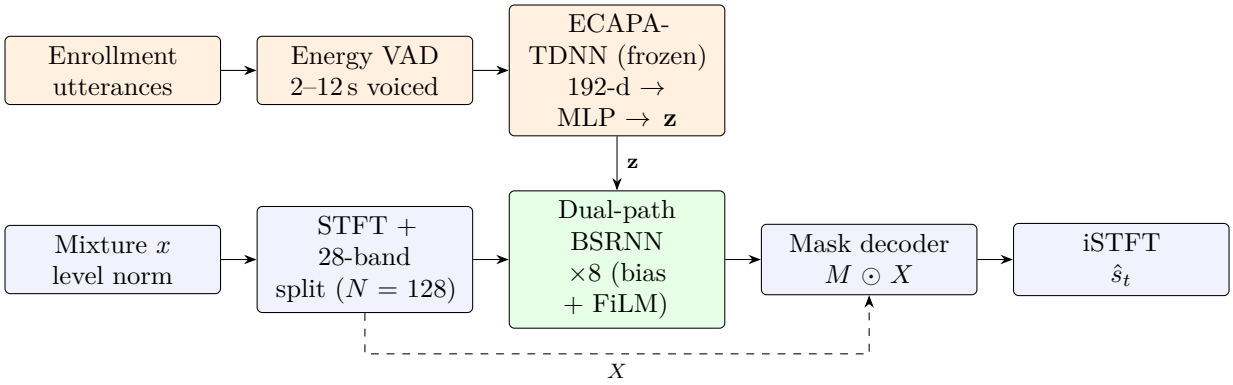


Figure 1: Speaker-conditioned BSRNN. The speaker vector $\mathbf{z}$ is added once to the band features and used by FiLM in every dual-path layer. The dashed path carries the mixture spectrogram to the mask multiplier.

Table 3: Speech corpus split and crop configuration

Role	LibriSpeech subset(s)	Speakers reported	Crop length
Training	`train-clean-100`, `train-clean-360`	251 + 921	4 s
Validation	`dev-clean`	40	6 s
Testing	`test-clean`	40	6 s

DEMAND [17] provides multichannel environmental recordings (kitchen, living room, park, street, metro, car, and so on). Sixteen of the 17 environments were available; `PCAFETER` was not available in the dataset mirror used. Each recording is split by time: the first 80% for training, the next 10% for validation and the last 10% for testing. No noise audio is shared across splits, but the same 16 environments appear in all three.

Table 4: Temporal partition of each DEMAND recording.

Role	Fraction of recording	Approximate interval	Use
Training	$[0.00, 0.80)$	First 80%	Random training noise crops
Validation	$[0.80, 0.90)$	Next 10%	Validation mixtures
Testing	$[0.90, 1.00]$	Final 10%	Fixed test mixtures

Training data are generated on the fly. Every call samples a fresh scene with one of four scenarios: target plus interferer plus noise (probability 0.45), target plus interferer (0.25), target plus noise (0.25), or target alone (0.05). The target-to-interferer ratio is uniform in $[-5, 5]$ dB and the speech-to-noise ratio is uniform in $[-5, 15]$ dB. With probability 0.5 the interferer covers only 30–100% of the clip at a random offset. The mixture level is drawn from $[-35, -15]$ dBFS and passed through a peak limiter at 0.99. Training clips last 4 s and evaluation clips 6 s. One model therefore learns target extraction, plain denoising and leaving clean speech untouched. No training example removes the target entirely (Section 8).

Table 5: Configured mixture categories. Probabilities sum to one.

Scenario key	Probability	Present components	Purpose
`speech_noise`	0.45	Target + interferer + noise	Jointly tests competing speech and environmental suppression.
`speech`	0.25	Target + interferer	Tests speaker selectivity without added environmental noise.
`noise`	0.25	Target + noise	Tests conventional noise suppression while preserving identity.
`clean`	0.05	Target only	Provides a clean reference/control; included in the supplied test aggregate.

For target RMS R_s, interfering-speaker RMS R_u, and noise power P_n, the source code scales the components to the sampled SIR and SNR approximately as

$$\alpha = \frac{R_s}{R_u + \epsilon}\, 10^{-\mathrm{SIR}/20}, \tag{1}$$

$$\beta = \sqrt{\frac{\mathbb{E}[(s + \alpha u)^2]}{\mathbb{E}[n^2] 10^{\mathrm{SNR}/10} + \epsilon}}. \tag{2}$$

If a component is absent, its scale is zero and the corresponding ratio is not applicable. The dataset sampler draws reproducible evaluation manifests using fixed validation and test seeds; validation uses 300 examples and testing uses 1,200 examples in the supplied configuration.

5.1 Front end and band split

Audio is processed at 16 kHz. The waveform is divided by its standard deviation, and the output is multiplied back, so the network sees level-independent input. An STFT with a 512-point Hann window (32 ms) and a 128-sample hop (8 ms) gives 257 complex bins per frame; a 4 s training clip yields 501 frames. STFT, iSTFT and the loss are computed in fp32 even under mixed precision.

The 257 bins are grouped into 28 non-uniform bands: 8 bands of 125 Hz below 1 kHz, 12 bands of 250 Hz between 1 and 4 kHz, and 8 bands of 500 Hz between 4 and 8 kHz. The last band takes the one leftover bin. For each band, the real and imaginary parts are layer-normalized and linearly projected to $N = 128$ features, producing a tensor of shape $[B, 28, T, N]$. The fine low-frequency resolution follows the harmonic structure of speech; the coarser high-frequency bands cost less and carry less formant information.

5.2 Speaker branch and conditioning

Enrollment audio is cleaned with an energy-based voice activity detector (25 ms frames, 10 ms hop, threshold 30 dB below the 95th-percentile frame energy, floor −55 dB, 10-frame hang). The same VAD is used when building the cached enrollment banks and in `embed_enrollment` for a user's own recording. Voiced speech from several of the speaker's utterances is concatenated and cut to 2, 3, 5, 8 or 12 seconds (probabilities 0.15, 0.20, 0.30, 0.20, 0.15), then passed through a frozen SpeechBrain ECAPA-TDNN (`spkrec-ecapa-voxceleb`) [7, 8]. The 192-d embedding is L2-normalized and mapped by a small MLP ($192 \to 256 \to 256$, PReLU) to a speaker vector $\mathbf{z}$.

The vector enters the separator in two places. First, it is linearly projected to N dimensions and added once to the band features before the first layer. Second, every dual-path layer computes a per-channel scale and shift,

$$[\boldsymbol{\gamma}, \boldsymbol{\beta}] = W\mathbf{z} + \mathbf{b}, \qquad \mathbf{h}' = \mathbf{h} \odot (1 + \boldsymbol{\gamma}) + \boldsymbol{\beta}, \tag{3}$$

where $W \in \mathbb{R}^{2N \times 256}$. The FiLM weights and bias are zero-initialized, so training starts from an unconditioned BSRNN and learns to use the speaker cue gradually.

All enrollment embeddings are pre-computed into banks (10 recipes per training speaker, 6 per validation and test speaker, about 11.7 k training embeddings), so the encoder costs no GPU time or memory during training. A mixture's own target utterance is never part of its enrollment.

5.3 Dual-path layers and mask decoder

Each of the eight layers applies FiLM, then a time block (LayerNorm, BiLSTM along T for every band, projection, residual) and a band block (the same structure along the 28 bands for every frame). The time block gives long-range temporal context; the band block models cross-frequency structure at the low cost of a 28-step sequence.

The decoder is applied per band: LayerNorm, Linear$(N, 4N)$, Tanh, Linear$(4N, 4 \cdot \text{bins})$ and a GLU [18], which produces a real and an imaginary mask value for every bin. The mask is unbounded and complex, so it can change phase as well as magnitude: $\hat{S}(t, f) = M(t, f)X(t, f)$.

With $N = 128$, LSTM hidden size 256 and 8 layers, the model has **16.83 M parameters** and needs 23.9 GMACs for a 6 s clip (Section 7.6). This is much larger than the compact CPU model targeted in the proposal; reducing it is part of the remaining work (Section 9).

6 Training Setup

6.1 Loss

For an estimate $\hat{s}$ and reference s, scale-invariant SDR [10] is

$$\text{SI-SDR}(\hat{s}, s) = 10 \log_{10} \frac{\|\alpha s\|^2}{\|\alpha s - \hat{s}\|^2}, \qquad \alpha = \frac{\langle \hat{s}, s \rangle}{\|s\|^2}, \tag{4}$$

computed after removing the mean of each signal. The training loss combines it with a multi-resolution STFT loss [11]:

$$\mathcal{L} = -\, w_1\, \mathbb{E}\big[\, \min(\text{SI-SDR},\, 30\,\text{dB})\big] + w_2\, \mathcal{L}_{\text{MR-STFT}}, \qquad w_1 = 1,\ w_2 = 5. \tag{5}$$

$\mathcal{L}_{\text{MR-STFT}}$ sums spectral-convergence and log-magnitude terms at three resolutions (window, hop) = (512,128), (1024,256) and (2048,512). The 30 dB clamp stops near-perfect examples, such as the clean scenario, from dominating the gradient. As a consistency check, a logged loss of about -5 with a training SI-SDR of about 11 dB implies an MR-STFT term of about 1.2.

6.2 Optimization

Table 6: Training configuration of the main run.

Item	Value
Model	BSRNN, $N = 128$, hidden 256, 8 layers, 28 bands (16.83 M parameters)
Optimizer	AdamW [19], weight decay 0.01
Learning rate	2000-step linear warm-up to 5×10^{-4}, cosine decay [20] to 10^{-5} over 100k steps
Batch	8 clips × 4 s
Epoch definition	1000 steps (data generated on the fly); 100 epochs = 100k steps = 800k scenes
Gradient clipping	global norm 5.0
Precision	fp16 autocast with GradScaler [21]; STFT, iSTFT and loss in fp32
Memory saving	Gradient checkpointing [22] on every dual-path layer
Safety	Non-finite-loss batches are skipped and counted
Checkpointing	`_last.pt` every epoch (model, optimizer, scaler, log); `_best.pt` on best validation SI-SDRi
Outcome	100 epochs completed; best validation SI-SDRi 11.84 dB at epoch 97
Hardware	NVIDIA L4 (Google Colab)
Speed	0.99 s/step, peak GPU memory 3.2 GiB, about 17.4 min/epoch (about 29 h for 100 epochs)

Training ran on Colab, so the pipeline is built to resume: manifests and embedding banks live on Google Drive, and training was resumed from the epoch-33 checkpoint. Figure 2 plots the run after that resume.

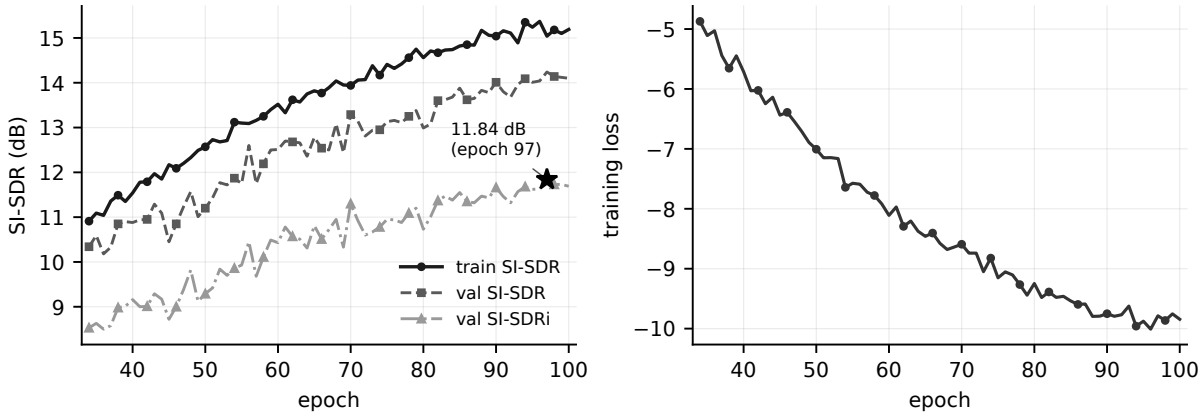


Figure 2: Training and validation SI-SDR, validation SI-SDRi (star: best epoch) and the combined loss for epochs 34–100.

7 Results

Training finished all 100 epochs. The best validation checkpoint (`tse_bsrnn_big_best.pt`) was reached at **epoch 97**, with a validation SI-SDRi of **11.84 dB** on unseen speakers. The fixed test set contains 1200 mixtures from 40 unseen speakers.

All results in this section, including the test-set tables, speaker-confusion analysis, WER check, qualitative examples and efficiency measurements, use this epoch-97 checkpoint.[1]

[1]The checkpoint file stores the zero-based epoch index 96, which corresponds to epoch 97 in the training log.

7.1 Training progress

After the resume at epoch 33 (validation SI-SDRi then 8.12 dB), validation SI-SDRi rose from 8.5 dB at epoch 34 to 11.7 dB at epoch 100, with a best value of 11.84 dB at epoch 97. The training loss fell from −4.9 to −9.8 and training SI-SDR from 10.9 to 15.2 dB. Over the last ten epochs the train–validation SI-SDR gap averaged about 1.1 dB, so there is no strong overfitting. The curve flattens as the learning rate anneals towards 10^{-5}: from epoch 90 onward, validation SI-SDRi stayed between 11.3 and 11.8 dB. No batches were skipped for non-finite loss in the final epochs. Swings of a few tenths of a dB from epoch to epoch are normal because the validation set has only 300 mixtures.

7.2 Test-set metrics

Table 7 gives SI-SDR, STOI and wideband PESQ on the test set with 95% bootstrap confidence intervals. The headline row group excludes the 65 clean-scenario items, where the mixture already equals the reference.

Table 7: Test metrics for the epoch-97 checkpoint, mean with 95% bootstrap CI. The headline column excludes the 65 clean-scenario items.

Metric	Non-clean (n=1135)	All items (n=1200)
SI-SDR input (dB)	0.28 [0.00, 0.56]	5.95 [4.62, 7.33]
SI-SDR output (dB)	11.92 [11.40, 12.44]	13.92 [13.22, 14.61]
SI-SDRi (dB)	**11.64 [11.17, 12.09]**	7.97 [6.99, 8.90]
STOI input	0.772 [0.765, 0.780]	0.785 [0.777, 0.792]
STOI output	0.882 [0.873, 0.891]	0.888 [0.879, 0.897]
PESQ input (wideband)	1.33	1.51
PESQ output (wideband)	2.38 [2.33, 2.42]	2.50 [2.45, 2.55]

On non-clean items the model improves SI-SDR by 11.6 dB, STOI by 0.11 (0.772 to 0.882) and PESQ from 1.33 to 2.38. The all-items SI-SDRi of 7.97 dB is lower purely because of the clean scenario (Table 8): there the input SI-SDR is about 105 dB, so even a 48.8 dB output SI-SDR registers as a −56 dB "improvement". That is a metric artifact rather than a quality problem. STOI is 1.00 on clean items and PESQ is 4.64 for the input and 4.62 for the output, so clean speech passes through almost unchanged.

Table 8: Results by scenario for the epoch-97 checkpoint. STOI and PESQ are reported as input/output.

Scenario	n	SI-SDR in	SI-SDR out	SI-SDRi [95% CI]	STOI in/out	PESQ in/out
noise	290	4.22	18.08	13.86 [13.17, 14.58]	0.90/0.95	1.75/2.99
speech	294	1.15	12.08	10.92 [9.93, 11.83]	0.76/0.90	1.28/2.54
speech_noise	551	−2.25	8.60	10.85 [10.15, 11.52]	0.71/0.84	1.14/1.97
clean	65	104.86	48.81	−56.05 [−58.36, −54.02]	1.00/1.00	4.64/4.62

Denoising alone (noise scenario) is easiest at 13.9 dB. Removing a competing talker (speech, 10.9 dB) and the hardest combination of talker plus noise (speech_noise, 10.9 dB) both give about 11 dB; the latter has the lowest output STOI (0.84) and PESQ (1.97). The SI-SDRi gain also shrinks as input SNR rises, from about 13 dB at low SNR to about 10 dB at high SNR, which is expected because there is less noise left to remove; absolute quality (PESQ) improves with SNR. The model does not collapse at hard ratios, and it still gains about 13 dB at SIR below −3.3 dB and at SNR below −1.7 dB. Across the 16 DEMAND environments the mean SI-SDRi ranges from 6.9 dB (restaurant, `PRESTO`) to 18.7 dB (washing, `DWASHING`); the hardest environments are crowded or transit scenes. The full SIR/SNR and environment breakdowns are in Section B.

7.3 Qualitative examples

Six test mixtures were drawn at random with a fixed seed and listened to. Table 9 lists SI-SDR before and after processing with the epoch-97 checkpoint; all six improved by 11.3 to 21.4 dB. Figure 3 shows the spectrograms for the best of the six. These examples are illustrative and do not replace the statistics above.

Table 9: Randomly chosen test examples (epoch-97 checkpoint).

Test index	Input SI-SDR (dB)	Output SI-SDR (dB)	Improvement (dB)
323	−3.50	12.27	+15.77
761	3.80	21.57	+17.77
611	2.52	17.88	+15.36
369	−0.77	20.67	+21.44
49	1.60	12.88	+11.27
1016	−0.66	11.38	+12.04

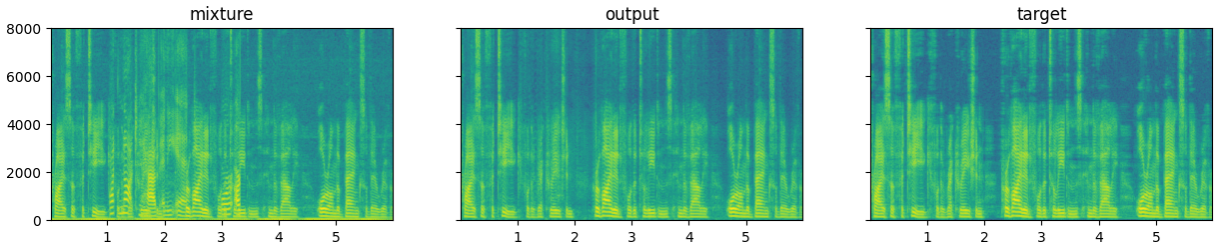


Figure 3: Test example 369 (+21.4 dB, the best of the six): mixture (left), model output (center) and clean target (right). The competing voice and broadband noise are removed, and the output closely resembles the target.

7.4 Speaker confusion

The most characteristic TSE failure is extracting the *wrong* speaker. For the 845 test mixtures that contain an interferer we embedded the output, the clean target and a 5 s segment of the interferer with ECAPA-TDNN, and counted an output as confused when it is closer in cosine similarity to the interferer than to the target. The confusion rate is **7.7%** (65 of 845). Mean similarity to the target rises from 0.584 for the mixture to 0.772 for the output. Figure 4 (top) shows the two clusters and (bottom) the rate by condition: it is roughly independent of SIR (6.5–8.3%), slightly higher with added noise (8.3% versus 6.5%), and depends on enrollment length (Table 10). Confusion is lowest with 12 s enrollments and highest with 3 s enrollments, so longer enrollment is likely to help; however, the trend is not strictly monotonic (2 s is below 3 s) and the 12 s bin contains only three confusions, so a controlled test is left for future work. All ten non-clean test mixtures with the lowest SI-SDRi are speaker confusions.

Table 10: Speaker-confusion rate by enrollment duration on the 845 interferer test mixtures.

Enrollment	n	Confused	Rate
2 s	121	10	8.3%
3 s	160	21	13.1%
5 s	234	18	7.7%
8 s	204	13	6.4%
12 s	126	3	2.4%

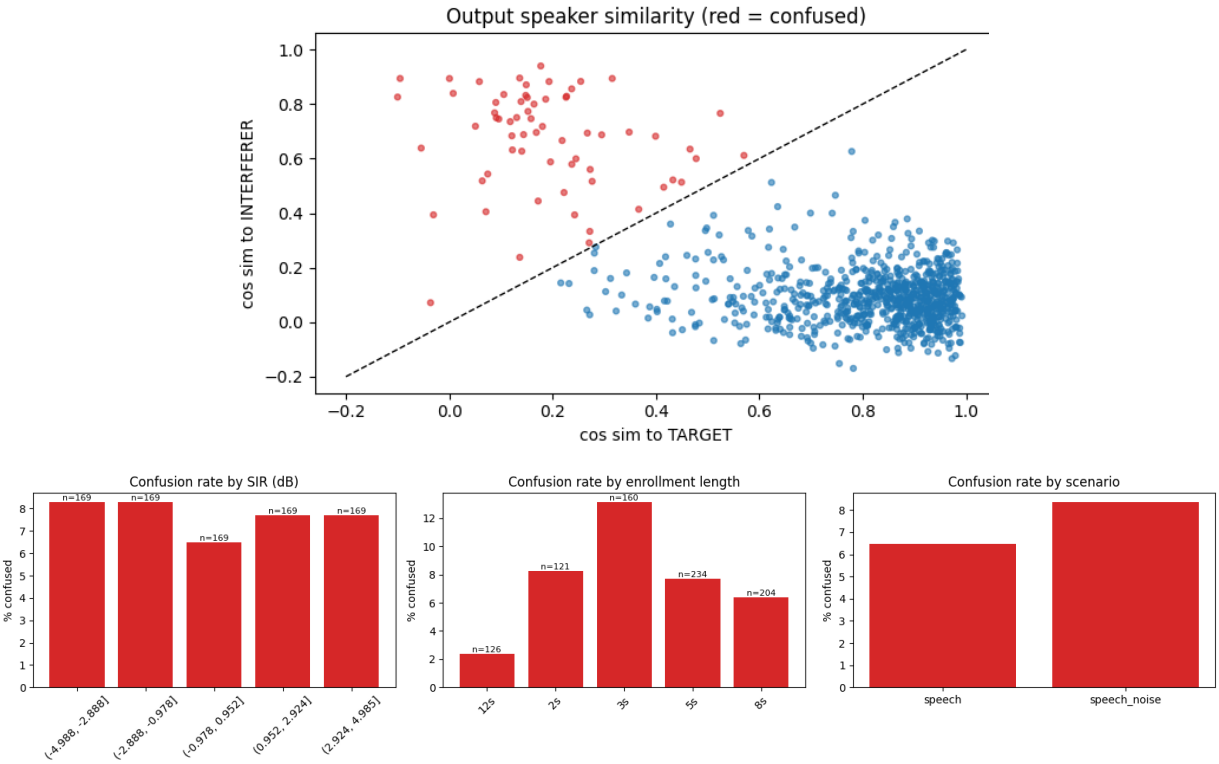


Figure 4: Top: output similarity to the target versus the interferer (red points are confused outputs). Bottom: confusion rate by SIR bin, enrollment length and scenario.

7.5 Downstream intelligibility

To measure usefulness beyond signal metrics, 300 random test mixtures were transcribed with Whisper `small.en` [23], using Whisper's transcript of the *clean target* as the reference rather than human transcripts, and WER was computed with `jiwer`. Table 11 shows that extraction more than halves the WER.

Table 11: Whisper WER (%) on 300 test mixtures from the epoch-97 checkpoint.

Subset	n	Mixture	Extracted
All	300	38.3	**18.1**
Excluding clean	285	40.2	**18.9**
noise	77	7.0	5.8
speech	77	40.4	10.5
speech_noise	131	60.0	31.9
clean	15	0.0	0.0

The largest gain is with a competing talker: WER drops from 40.4% to 10.5% for speech and from 60.0% to 31.9% for speech plus noise. Noise-only mixtures are already easy for Whisper (7.0%), and clean speech is not damaged (0% WER).

Additional diagnostics use the epoch-97 checkpoint. It improves SI-SDR on 93.7% of the 1135 non-clean test mixtures (88.6% of all 1200, since the 65 clean items cannot improve). On non-clean items its SI-SDRi correlates weakly and negatively with SNR ($r = -0.19$) and SIR ($r = -0.13$) and is essentially uncorrelated with input SI-SDR ($r = -0.05$) and enrollment duration ($r = 0.05$). For the 845 mixtures with an interferer, the mean SI-SDRi is 12.84 dB when the output matches the target speaker and -12.67 dB when it does not; if the confused outputs behaved like the correct ones, the non-clean SI-SDRi would rise from 11.64 dB to 13.10 dB. The relative WER

reduction is 52.9%.

7.6 Efficiency

Table 12 reports inference cost on an NVIDIA L4 for a 6 s clip, processed offline in one pass.

Table 12: Model size and inference cost for a 6 s clip, batch 1, on an NVIDIA L4.

	fp32	fp16 autocast
Parameters	16.83 M	
Compute	23.87 GMACs per clip (3.98 GMACs per second of audio)	
Time per clip	164.9 ms	181.1 ms
Offline RTF	0.0275	0.0302
Peak GPU memory	1057 MiB	870 MiB

The fp32 pass takes 164.9 ms for this 6 s clip; fp16 saves memory but is slower at batch size one. These are offline GPU measurements. They do not imply low latency: the bidirectional LSTMs need the whole clip, and CPU speed has not yet been measured.

8 Challenges and Limitations

- **Non-causal model.** Bidirectional LSTMs use future frames, so the current model cannot run with low latency. This departs from the proposal's low-latency goal, and a causal variant has not been built.
- **Size and compute.** At 16.83 M parameters and about 4 GMACs per second of audio, the model is not yet compatible with the compact CPU target. CPU speed is unmeasured, and the reported RTF is an offline GPU figure.
- **Speaker confusion.** In 7.7% of interferer mixtures the output matches the interferer better than the target, mostly with 2–3 s enrollments (Section 7.4). Target-absent behavior is untested, so the system may extract the wrong voice when the target is silent.
- **No unseen-noise test.** DEMAND recordings are split by time, so the same 16 environments appear in training, validation and test. Performance on unseen noise types is therefore not demonstrated. Reverberation is also absent from both training and test.
- **Enrollment diversity.** Enrollments are VAD-cleaned from clean recordings, with only 10 recipes per training speaker. Real enrollments are noisier. The same energy VAD is used for the cached banks and for a user's own recording, as described in Section 4.
- **Length mismatch.** Training uses 4 s clips and evaluation 6 s clips. Long inputs are processed in one pass or in 30 s chunks without overlap-add, which may cause boundary artifacts.
- **Evaluation status.** The 300-mixture validation set is small. The clean scenario distorts the aggregate SI-SDRi, so non-clean results are the headline.
- **WER reference.** The WER check uses Whisper transcripts of the clean target as the reference, not human transcripts, and covers 300 mixtures.
- **Reproducibility.** The configured random seed is not applied inside the training loop, and dataset workers use unseeded generators, so reruns see different sampled mixtures.

9 Remaining Work and Revised Plan

Table 13 lists the remaining tasks with a primary owner and a priority. Two items are high priority: the conditioning ablation promised in the proposal and a test on unseen noise environments. CPU efficiency and a causal variant are medium priority and depend on the time left after the high-priority items.

Table 13: Remaining work and primary owner. Ownership follows the proposal's module assignments.

Task	Description	Owner	Priority
Conditioning ablation	Compare additive-only, FiLM-only and concatenation/gating, as promised in the proposal	Prashamsa	High
Unseen-noise test	Hold out whole DEMAND environments, or add MUSAN or self-recorded noise, for testing	Austina	High
Efficiency measurement	RTF, latency and memory on CPU (GPU RTF 0.028 already measured)	Sostika	Medium
Model compression and causal variant	Smaller configuration; unidirectional LSTM and causal STFT if time permits	Prashamsa, Sostika	Medium
Generalization additions	Reverberation (RIRs), noisy and longer enrollments to cut speaker confusion, target-absent samples	Austina	Medium
Demo and documentation	Packaged inference script or small demo, final report and presentation	All	High

10 Conclusion

We moved from the proposal's causal Conv-TasNet design to a speaker-conditioned BSRNN, and training is complete. The model reached a validation SI-SDRi of 11.84 dB on unseen speakers. On the test set, non-clean SI-SDR improved by about 11.6 dB, STOI rose from 0.77 to 0.88 and PESQ from 1.33 to 2.38; the Whisper WER check fell from 38.3% to 18.1%. The trade-off is clear: the current model is larger and non-causal, and 7.7% of interferer mixtures were classified as speaker confusions. We still need the conditioning ablation, an unseen-noise test and CPU measurements. On the team side, Austina led data preparation, Prashamsa led the speaker encoder and conditioning work, and Sostika led training and inference; all three shared the literature review, evaluation and writing.

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A Reproducibility Notes

- **Result files:** `test_metrics.csv` (per-mixture SI-SDR, STOI and output PESQ), `pesq_in.csv` (input PESQ), `speaker_confusion.csv` and `wer_transcripts.csv`, all generated with the epoch-97 checkpoint.
- **Code:** `tselib.py` contains the model, sampler, datasets, loss and training loop, with two YAML configuration files. These are kept on Google Drive under `tse-project/`. The project repository is at `https://github.com/frecklehead/SIC-AI-Project`.
- **Training notebook:** downloads LibriSpeech and DEMAND, loads manifests and cached embedding banks, patches gradient checkpointing, and calls `T.train(...)`.
- **Test notebook:** loads the best checkpoint, plays the mixture, output and target, computes SI-SDR, STOI and PESQ on the test manifest, and separates user-supplied files.
- **Software:** PyTorch 2.x (CUDA), SpeechBrain, `pystoi`, `pesq`, `jiwer` and Whisper through `transformers` for the WER check.
- **Analysis cells:** per-scenario, SNR, SIR and noise-environment breakdowns, bootstrap confidence intervals, speaker-confusion analysis, WER and FLOP/RTF benchmarking are in the training notebook after the test-loading cell.

B Additional Results

Table 14: SI-SDRi (dB) by difficulty. SIR bins use the 845 mixtures with an interferer; SNR bins use the 841 mixtures with noise.

SIR bin (dB)	n	SI-SDRi	PESQ	SNR bin (dB)	n	SI-SDRi	PESQ
$[-5.0, -3.3]$	131	12.57	1.96	$[-5.0, -1.7]$	155	13.24	1.86
$(-3.3, -1.7]$	152	10.84	2.00	$(-1.7, 1.7]$	136	13.68	2.19
$(-1.7, 0.0]$	134	12.23	2.14	$(1.7, 5.0]$	142	12.57	2.38
$(0.0, 1.7]$	147	10.95	2.20	$(5.0, 8.3]$	135	11.63	2.47
$(1.7, 3.3]$	144	9.32	2.28	$(8.3, 11.6]$	126	10.38	2.48
$(3.3, 5.0]$	137	9.52	2.41	$(11.6, 14.9]$	147	9.67	2.59

Table 15: Mean SI-SDRi and output quality per DEMAND environment for the epoch-97 checkpoint (all 841 test mixtures that contain noise).

Env.	n	SI-SDRi	STOI	PESQ	Env.	n	SI-SDRi	STOI	PESQ
DKITCHEN	57	14.95	0.93	2.74	OOFFICE	48	13.44	0.89	2.49
DLIVING	61	9.50	0.85	1.92	PRESTO	49	6.91	0.79	1.74
DWASHING	33	18.72	0.93	2.96	PSTATION	64	8.38	0.83	2.06
NFIELD	44	13.88	0.90	2.39	SPSQUARE	49	11.13	0.87	2.14
NPARK	52	11.74	0.85	2.05	STRAFFIC	51	11.25	0.90	2.43
NRIVER	44	12.90	0.87	2.27	TBUS	63	14.46	0.93	2.80
OHALLWAY	53	13.94	0.89	2.49	TCAR	52	15.03	0.94	3.03
OMEETING	65	9.20	0.84	1.98	TMETRO	56	8.78	0.83	1.94

Table 16: Conv-TasNet baseline configuration and result. The saved best checkpoint is epoch 21.

Configuration	Causal; $N = 512$, $H = 512$; film2 conditioning; 3 s segments; batch 3 with gradient accumulation 2; learning rate 10^{-3}; 80 epochs planned
Best checkpoint	Epoch 21, validation SI-SDRi 5.90 dB
20-clip check	3.88 dB on training voices; 3.60 dB on held-out voices and noise
Decision	Dropped after BSRNN reached 11.84 dB validation SI-SDRi